

Types of Prompts in LLM's

Introduction

Prompts are the instructions given to a Large Language Model (LLM) to get a response. Different types of prompts provide different levels of understandings to the model. There are six main types of prompts

1 Zero-Shot Prompting

In zero-shot prompting, we directly ask the LLM to perform a task without providing any example. The model understands the task based only on the instruction given.

Example:

“Translate this sentence into French: I love cats.”

Output: J'aime les chats

2 One-Shot Prompting

In one-shot prompting, we provide the LLM with one example of how we want the task to be performed.

Example:

Cat → Animal

Rose → ?

Output: Rose → Flower

3 Few-Shot Prompting

In few-shot prompting, we provide two or more examples before asking the LLM to perform the actual task. These examples help the model understand the pattern more clearly.

Example:

Happy → Positive

Sad → Negative

Angry → ?

Output: Negative

4 Chain-of-Thought (CoT) Prompting

In Chain-of-Thought prompting we ask the LLM to work through a problem step by step. The problem is broken into smaller parts to help the model solve reasoning-based tasks.

Example:

“Solve 25×12 step by step.”

Output: 25×12

Step 1: Break 12 into $10 + 2$

$$25 \times 12 = 25 \times (10 + 2)$$

Step 2: Multiply separately

$$25 \times 10 = 250$$

$$25 \times 2 = 50$$

Step 3: Add the results

$$250 + 50 = 300$$

Answer: 300

5 Role-Based Prompting

In role-based prompting, we tell the LLM to act as a particular person or professional. This helps the model respond from the perspective of that role.

Example:

“You are an experienced Python teacher. Explain loops to a beginner.”

Output: Python if Statement — Beginner Explanation

An if statement is used to make a decision in Python.

It means:

”If this condition is true, then do this.”

Basic Syntax if condition: code to execute Simple Example `age = 18`

`if age >= 18: print("You are an adult.")` How it works `age = 18` stores the value 18.

Python checks the condition:

`age = 18`

Since `18 >= 18` is True, Python executes:

`print("You are an adult.")`

Output:

You are an adult. if with else

When you want to do something different if the condition is false, use else:

`age = 15`

`if age >= 18: print("You are an adult.") else: print("You are a minor.")`

Output:

You are a minor. Important: Indentation

Python uses indentation (spaces) to know which code belongs to the if.

Correct:

```
if age >= 18: print("Adult")
```

Incorrect:

```
if age >= 18: print("Adult")
```

Usually, we use 4 spaces for indentation.

One more example marks = 75

```
if marks >= 40: print("Pass") else: print("Fail")
```

Here Python asks:

"Are marks greater than or equal to 40?"

Since 75 >= 40 is true, the output is:

Pass

In short: if helps Python make decisions based on conditions.

6 Instruction-Based Prompting

In instruction-based prompting, we give the LLM clear and specific instructions about what we want. We can also specify the required format, length, or style of the response.

Example:

"Summarize the water cycle in 3 bullet points using simple language."

Output:article

Water Cycle

- **Evaporation:** The Sun heats water from oceans, lakes, and rivers, turning it into water vapor.
- **Condensation:** Water vapor rises, cools, and forms clouds.
- **Precipitation:** Water falls from clouds as rain, snow, or hail and returns to Earth.